



BITS Pilani
Pilani Campus

Module 2: Mechanical Separations (Chapter 29)

Lecture 8

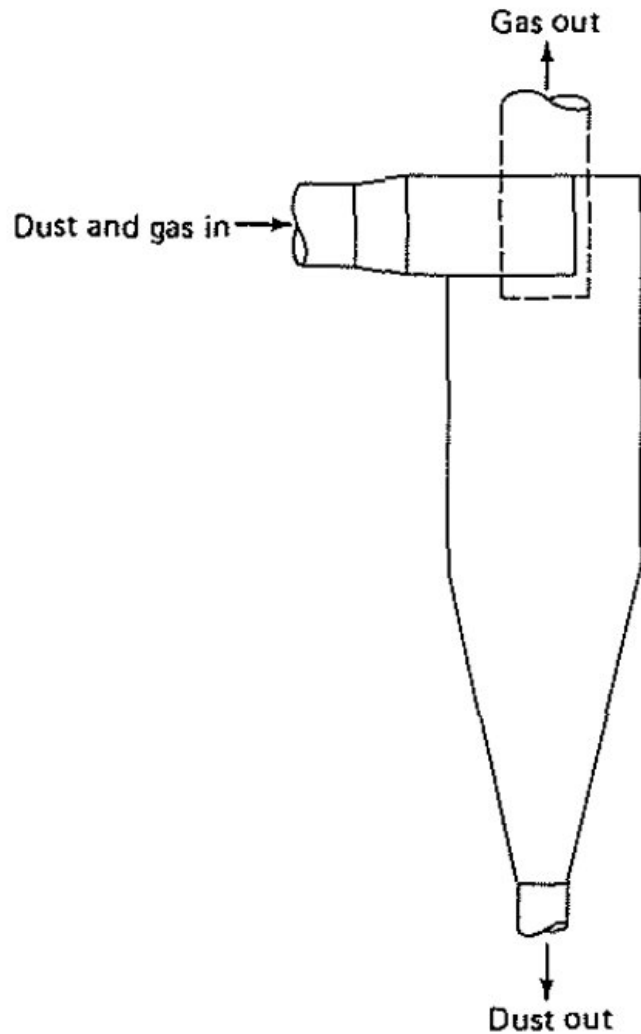
Somak Chatterjee



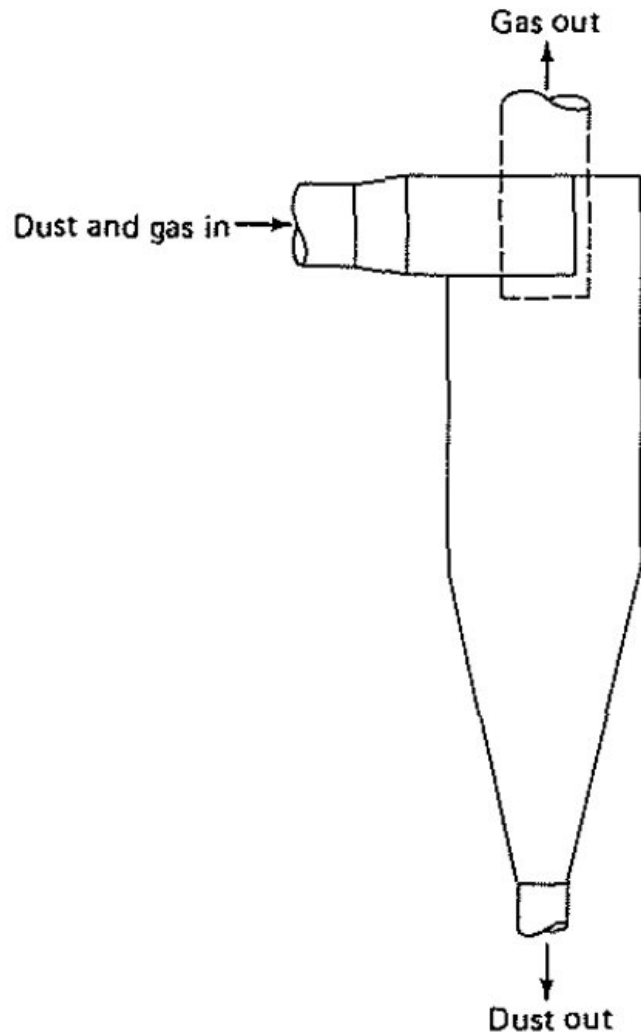
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Centrifugal Sedimentation Process



- Most centrifugal separators for removing particles from gas streams contain no moving parts.
- It consists of a vertical cylinder with a conical bottom, a tangential inlet near the top, and an outlet for dust at the bottom of the cone.
- The inlet is usually rectangular.
- The outlet pipe is extended into the cylinder to prevent short-circuiting of air from inlet to outlet.



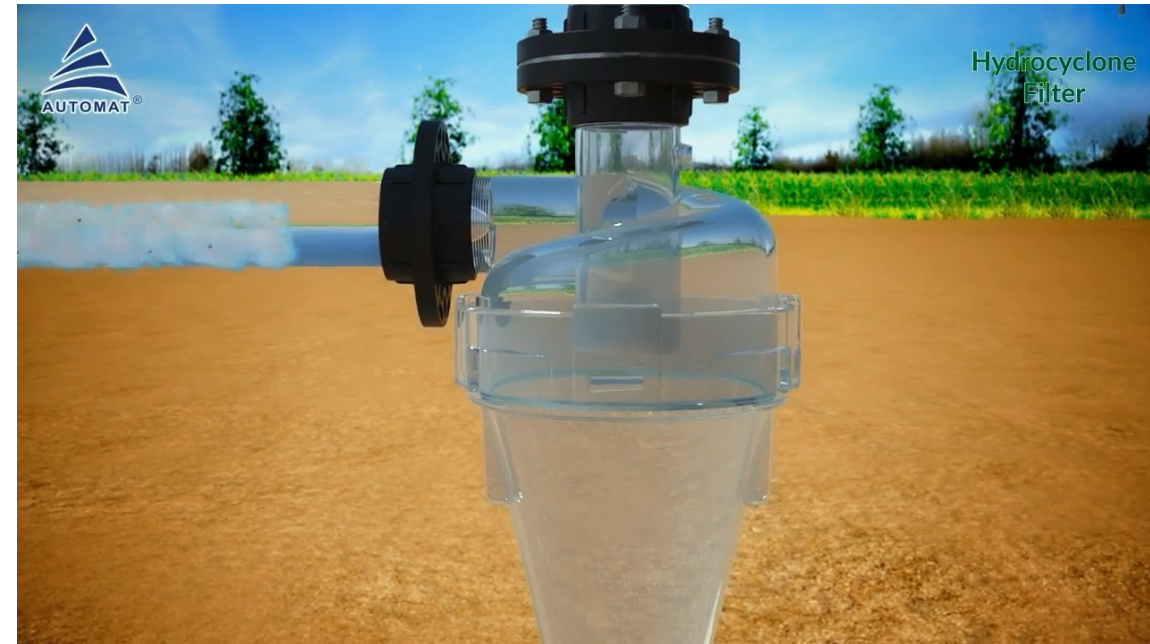
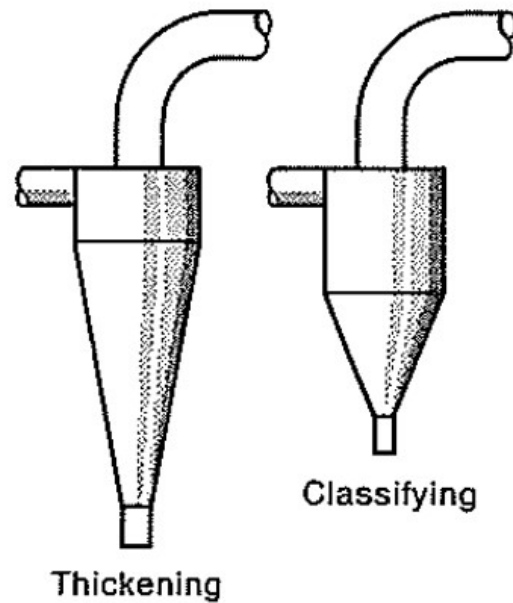
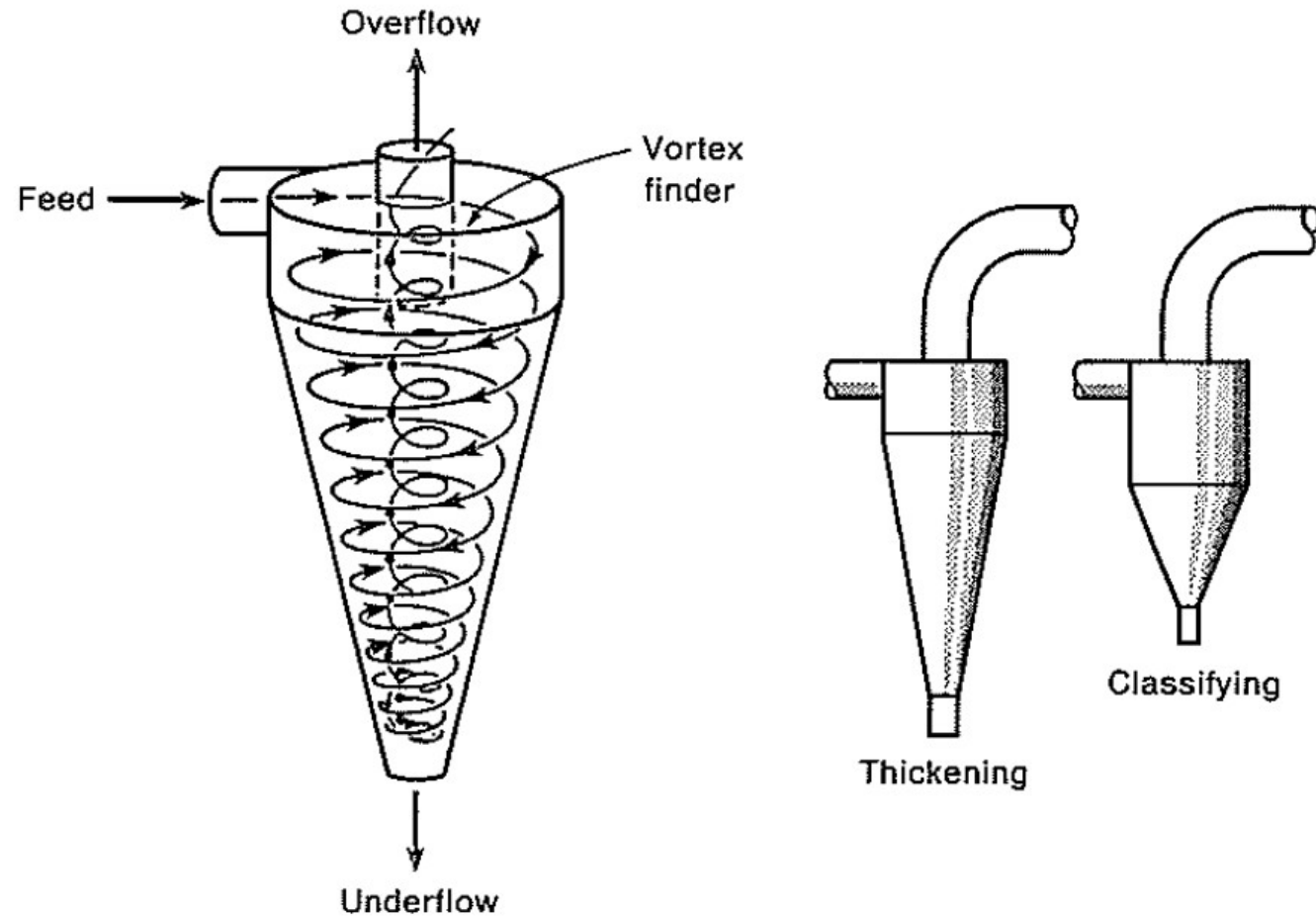
- The incoming dust-laden air travels in a spiral path around and down the cylindrical body of the cyclone.
- The centrifugal force developed in the vortex tends to move the particles radially toward the wall, and the particles that reach the wall slide down into the cone and are collected.
- The cyclone is basically a settling device in which a strong centrifugal force, acting radially, is used in place of a relatively weak gravitational force acting vertically.

- The centrifugal force F_c at radius r is equal to mu_{tan}^2 / r
 - where m is the mass of the particle and
 - u_{tan} is its tangential velocity.
- The ratio of the centrifugal force to the force of gravity is then

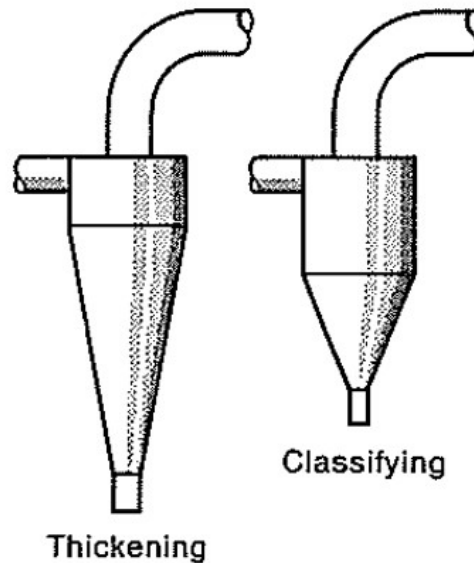
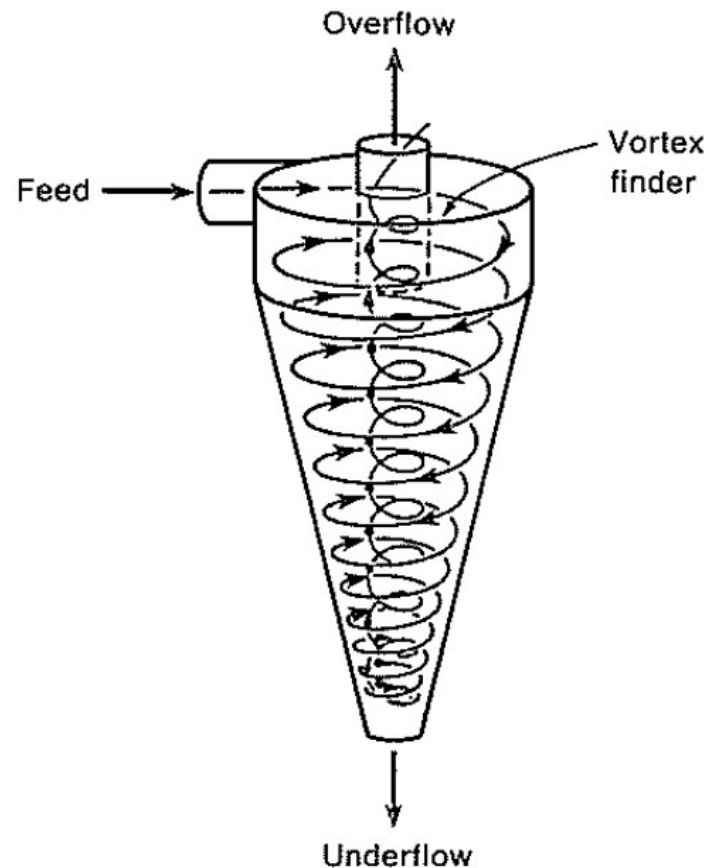
$$\frac{F_c}{F_g} = \frac{mu_{tan}^2 / r}{mg}$$

- This is called as separation factor

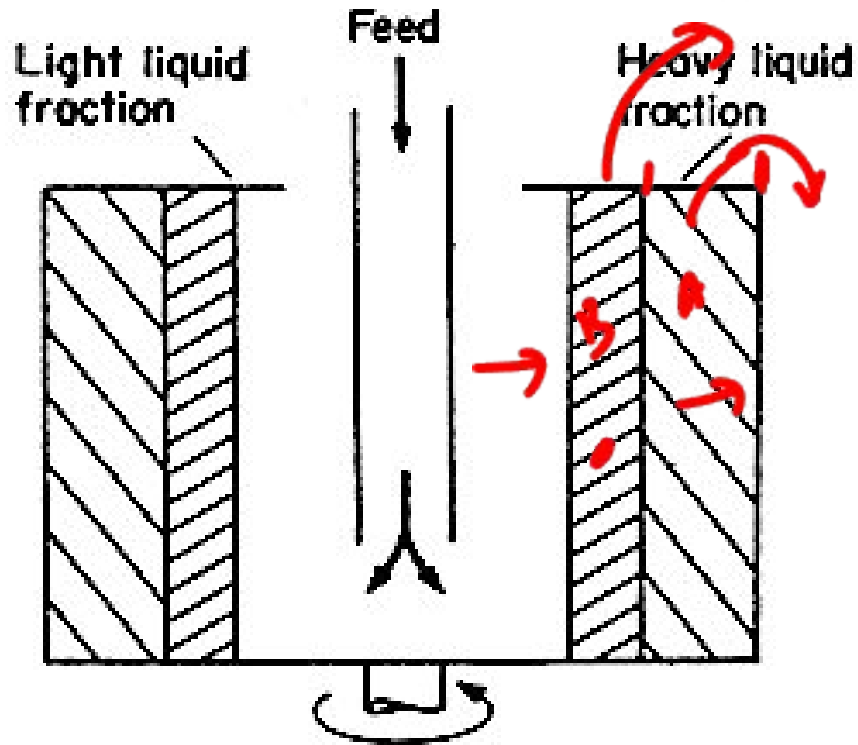
Hydroclones



https://www.youtube.com/watch?v=8PJfFF_HmUo



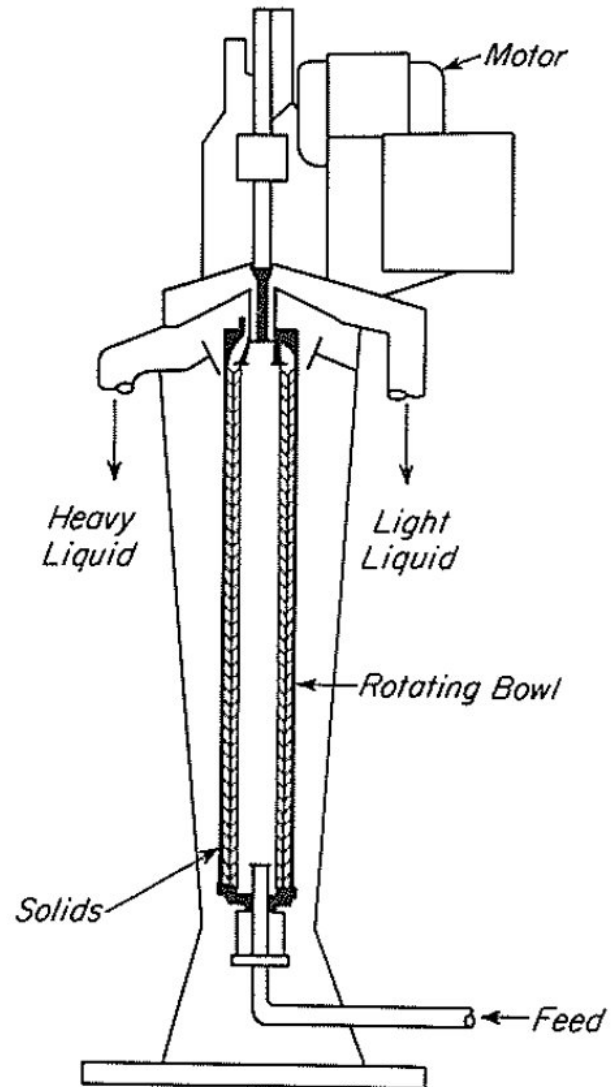
- Cyclones are also used for separating solids from liquids, sometimes as thickeners but much more commonly as classifiers.
- In these services they are called hydrocyclones or hydroclones.
- Feed enters tangentially at high velocity near the top.
- The liquid follows a spiral path near the vessel wall, forming a strong downward vortex.
- Large or heavy solid particles separate to the wall and are pushed downward and out of the cyclone as a slurry or paste.
- A variable-discharge orifice controls the consistency of the underflow.
- Most of the liquid goes back upward in an inner vortex and leaves through the central discharge pipe, which is known as a vortex finder.



- Immiscible liquids are separated industrially in centrifugal decanters. ✓
- The separating force is much larger than that of gravity, and it acts in the direction away from the axis of rotation instead of downward toward the earth's surface.
- The main types of centrifugal decanters are tubular centrifuges and disk centrifuges.

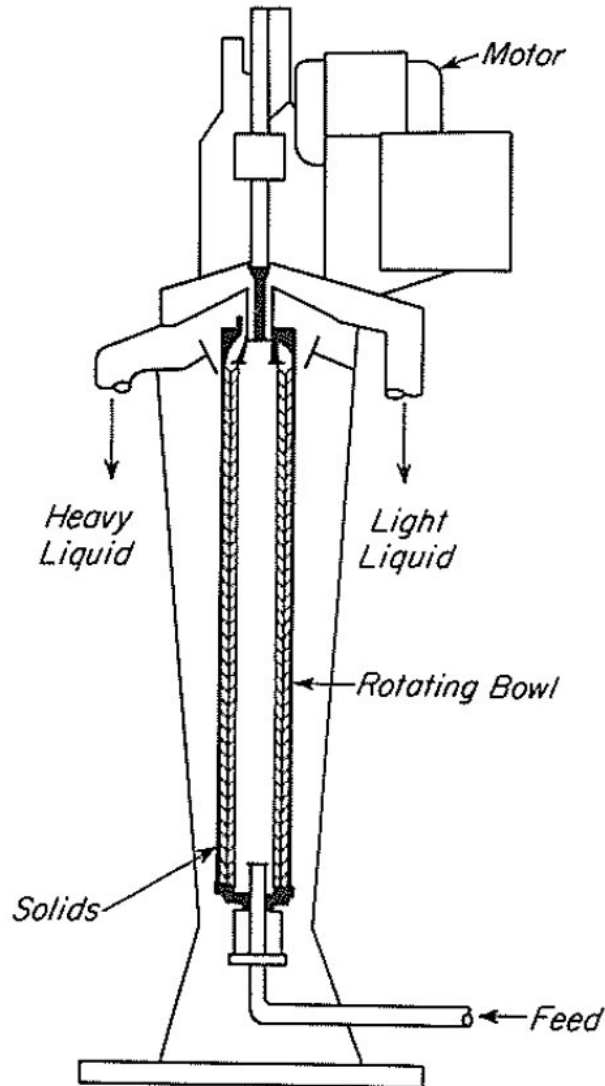
Perry's Handbook

Centrifugal Decanters: Tubular Centrifuge



<https://www.youtube.com/watch?v=wg3xrDbyi4o>

Centrifugal Decanters: Tubular Centrifuge



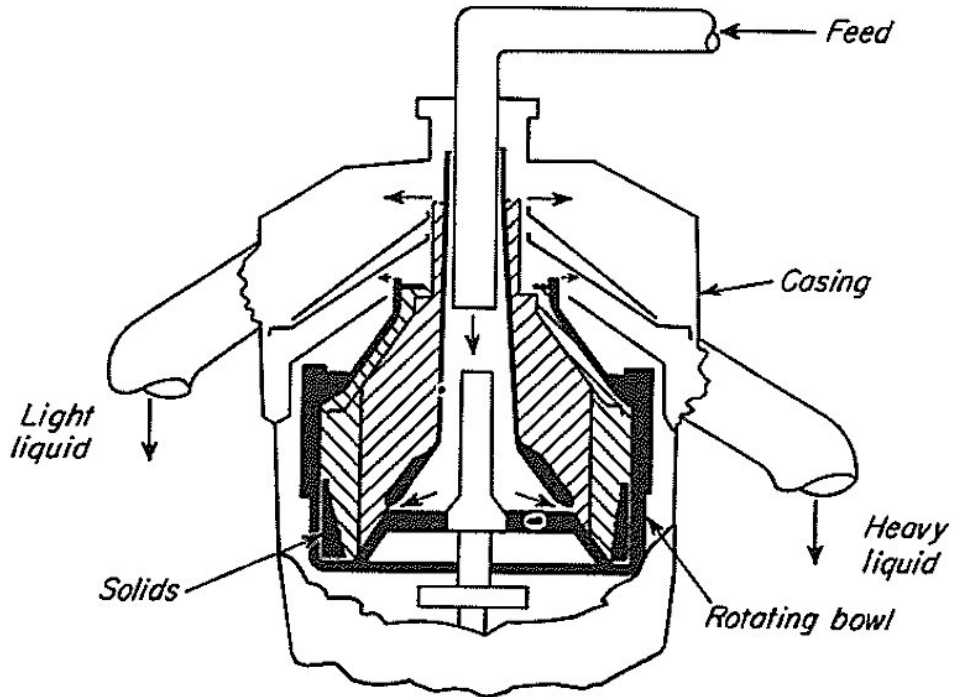
- Feed enters from a stationary nozzle inserted through an opening in the bottom of the bowl.
- It separates into two concentric layers of liquid inside the bowl.
- The inner, or lighter layer spills over a weir at the top of the bowl; it is thrown outward into a stationary discharge cover and from there to a spout.
- Heavy liquid flows over another weir into a separate cover and discharge spout.
- The weir over which the heavy liquid flows is removable and may be replaced with another having an opening of a different size.

Centrifugal Decanters: Disk Centrifuge

innovate

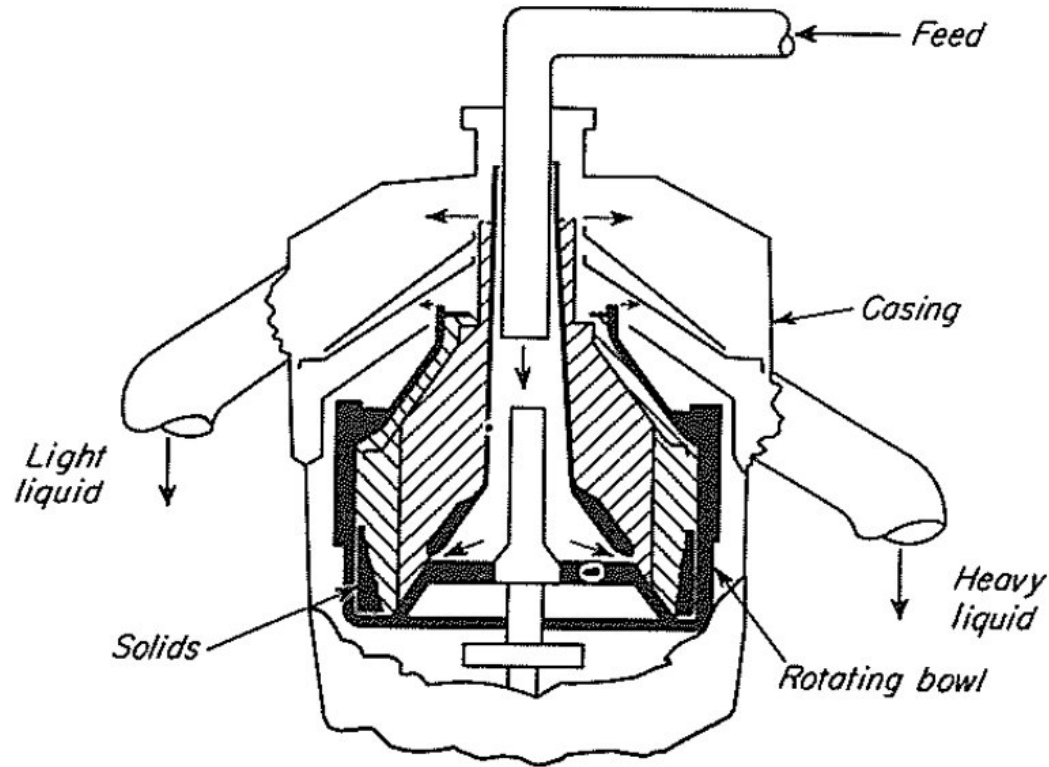
achieve

lead



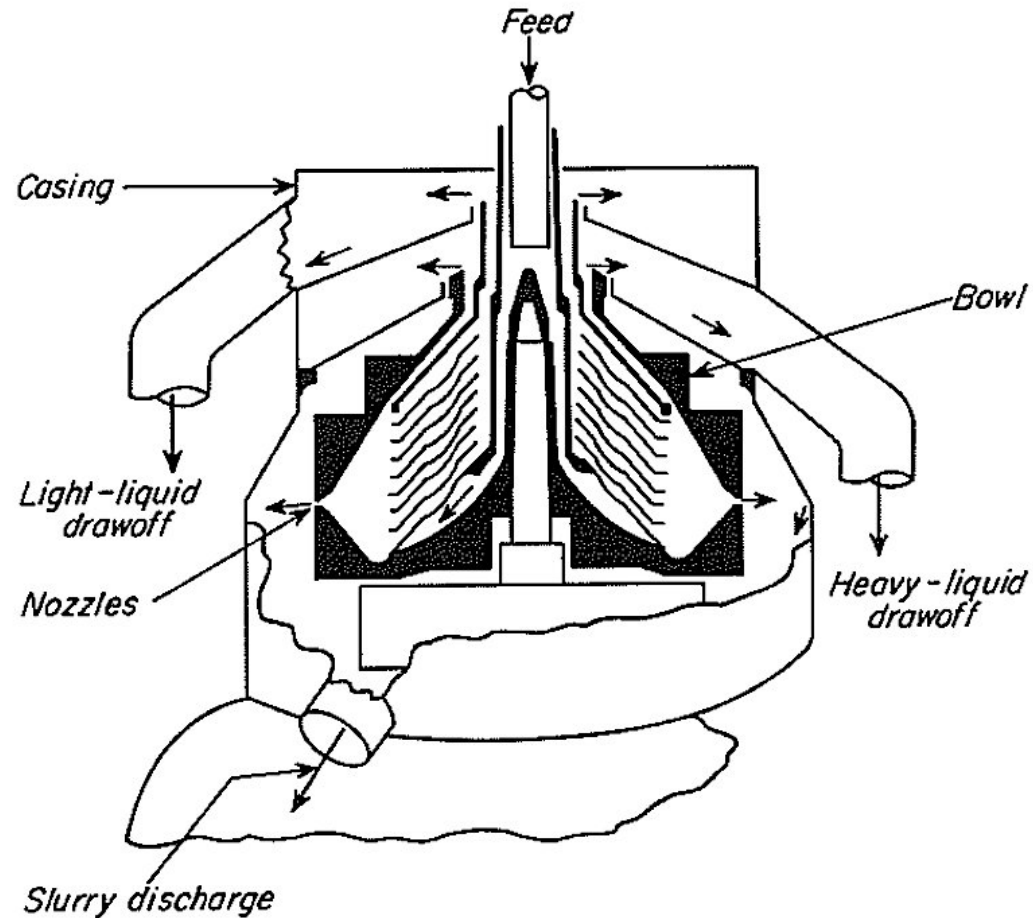
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Centrifugal Decanters: Disk Centrifuge



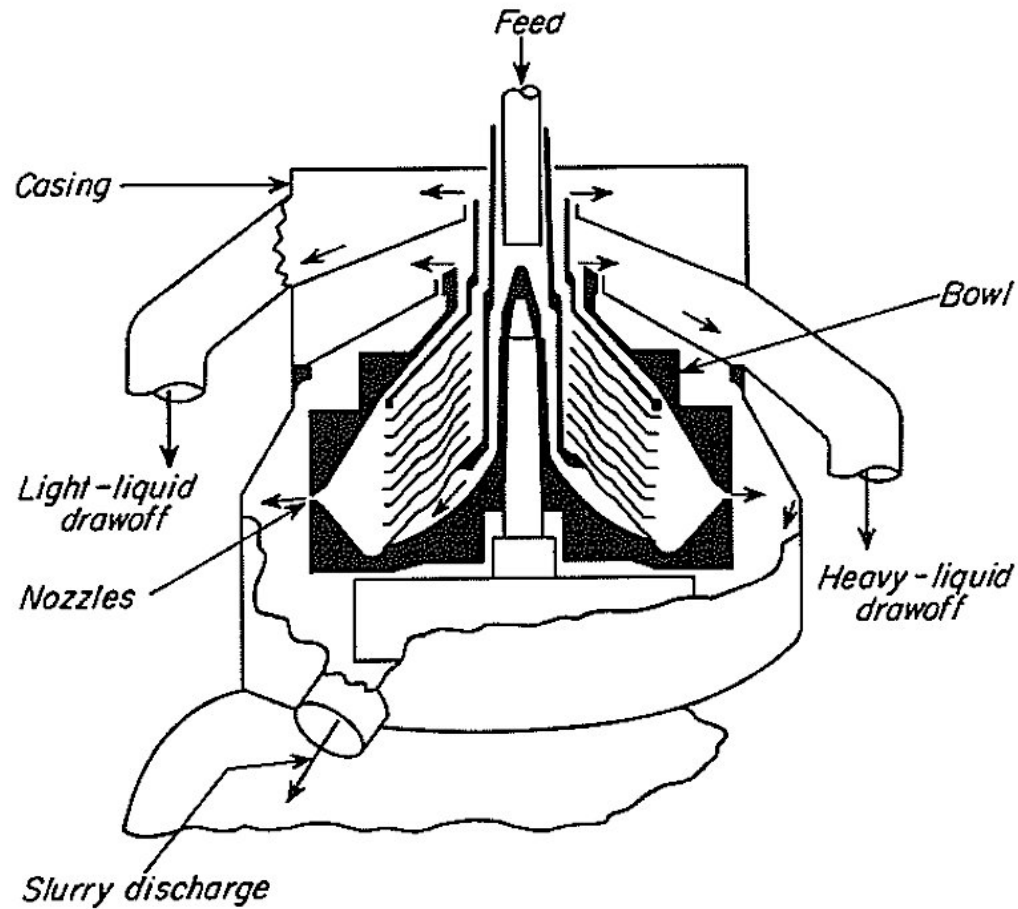
- Feed enters from above through a stationary pipe set into the neck of the bowl.
- Two liquid layers are formed as in a tubular centrifuge; they flow over adjustable dams into separate discharge spouts.
- Inside the bowl and rotating with it are closely spaced "disks," which are actually cones of sheet metal set one above the other.
- Matching holes in the disks about halfway between the axis and the wall of the bowl form channels through which the liquids pass.
- In operation feed liquid enters the bowl at the bottom, flows into the channels, and upward past the disks.
- Heavier liquid is thrown outward, displacing lighter liquid toward the center of the bowl.
- In its travel the heavy liquid very soon strikes the underside of a disk and flows beneath it to the periphery of the bowl without encountering any more light liquid.
- Light liquid similarly flows inward and upward over the upper surfaces of the disks.
- Since the disks are closely spaced, the distance a drop of either liquid must travel to escape from the other phase is short, much shorter than in the comparatively thick liquid layers in a tubular centrifuge.
- Disk centrifuges are particularly valuable where the purpose of the centrifuging is not complete separation but the concentration of one fluid phase, as in the separation of cream from milk and the concentration of rubber latex.

Centrifugal Decanters: Nozzle Discharge Centrifuge

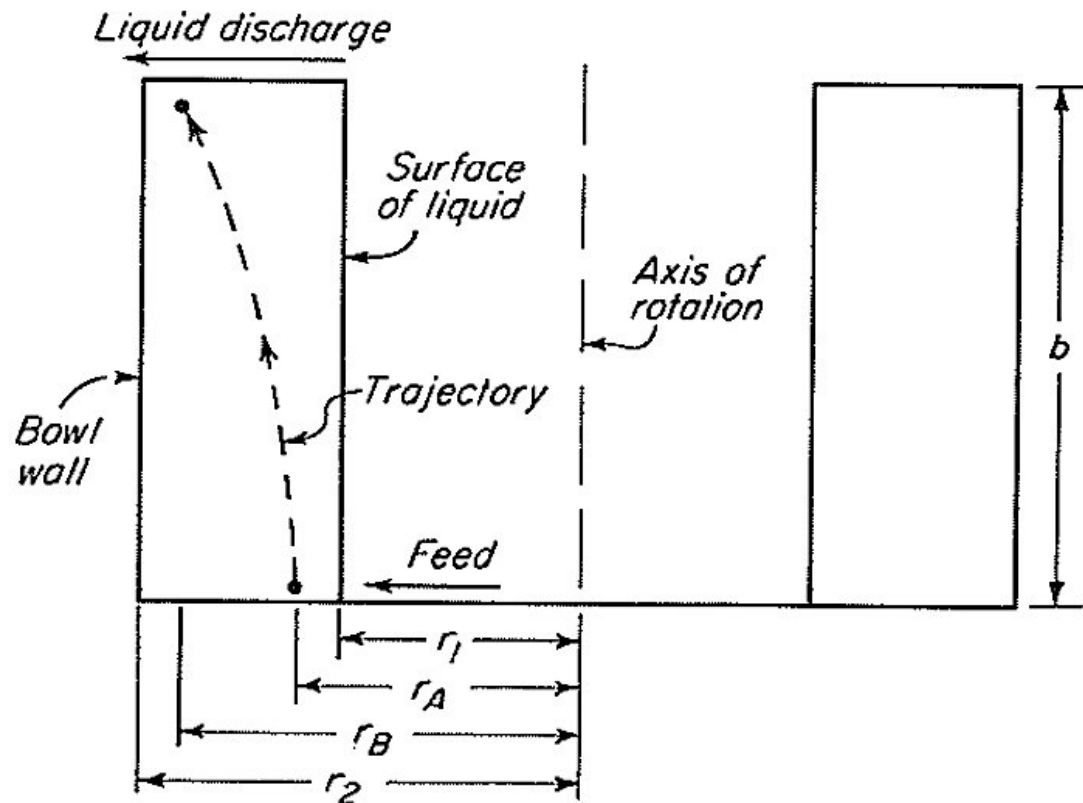


<https://www.youtube.com/watch?v=PPGke-SwpP4>

Centrifugal Decanters: Nozzle Discharge Centrifuge



- This separator is a modified disk-type centrifuge with a double conical bowl.
- In the periphery of the bowl at its maximum diameter is a set of small holes, or nozzles, perhaps 3 mm in diameter.
- The central part of the bowl operates in the same way as the usual disk centrifuge, overflowing either one or two streams of clarified liquid.
- Solids are thrown to the periphery of the bowl and escape continuously through the nozzles, together with considerable liquid.



Principle of Centrifugal Sedimentation

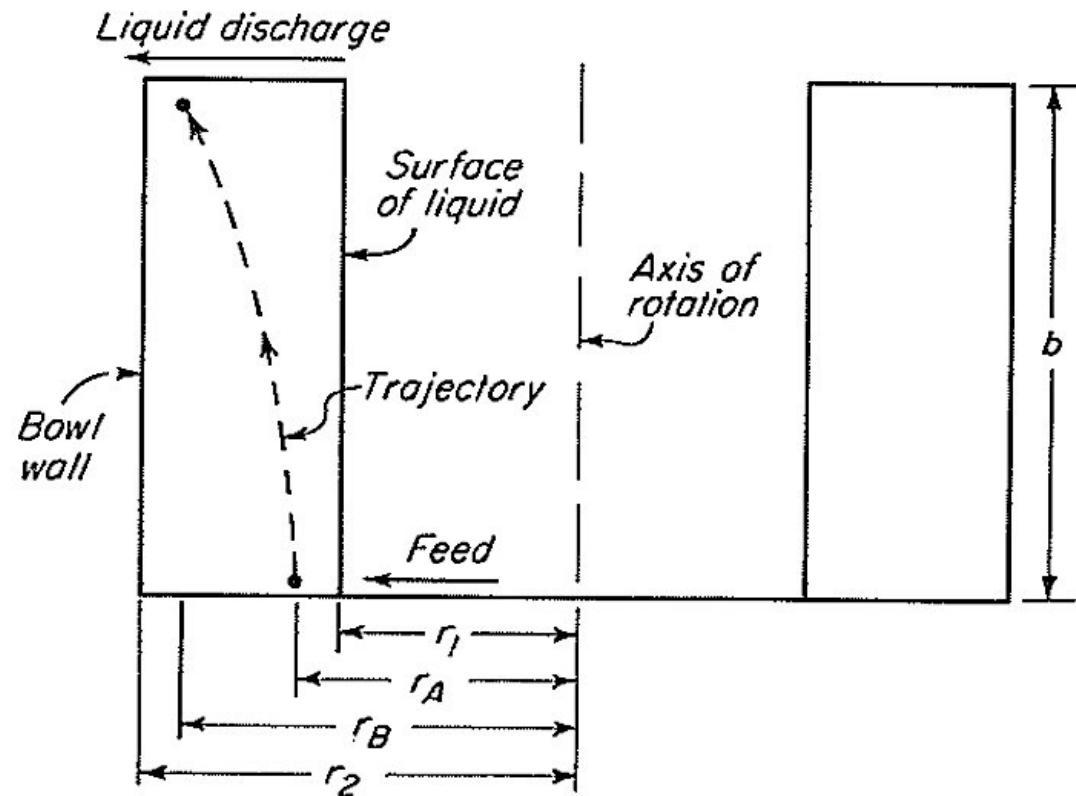
→ Particle of Given Size is removed, when particle reaches wall of Separator bowl.

→ Particle moves radially with terminal Velocity

→ Diameter of Smallest particle, that is removed is Calculated

Feed point at bottom
liquid Discharge at top

liquid moves upward through the bowl at constant Velocity carrying Solid Particles with it.



Particle settle at bottom of the bowl at same position in the liquid $\rightarrow r_A$ from axis of rotation

Settling time limited by residence time of liquid in the bowl

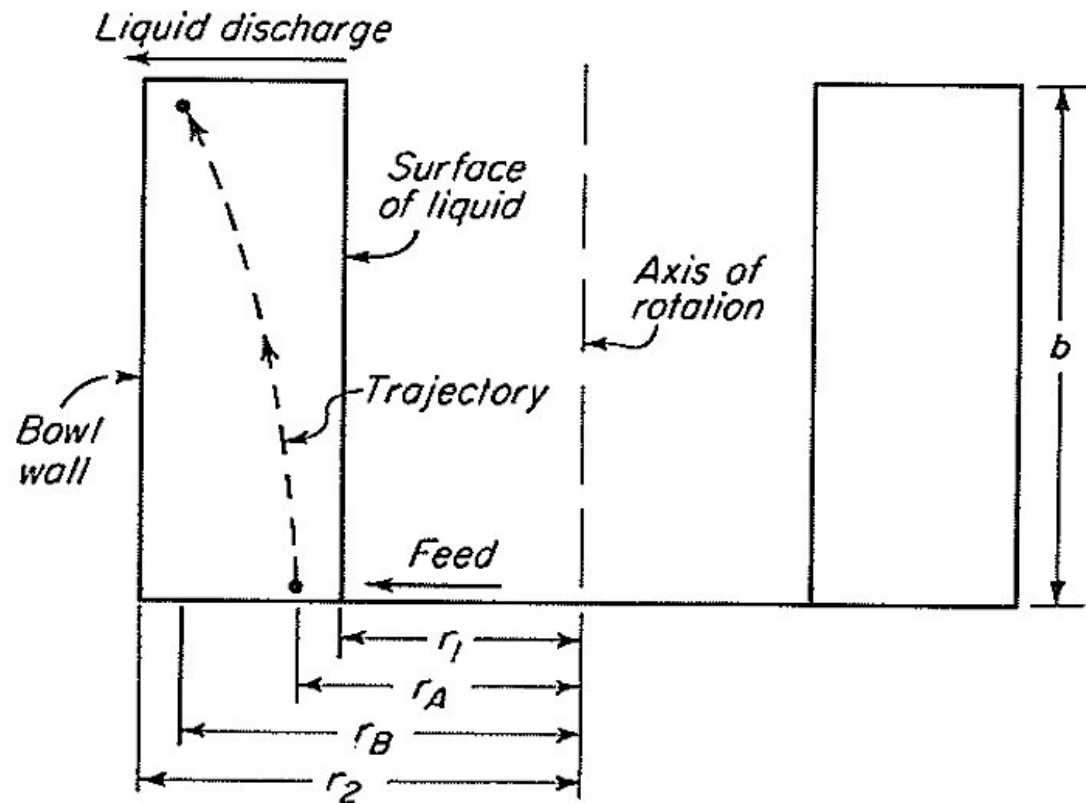
At the end of settling time, particle is " r_B " from axis of rotation.

$r_1 \rightarrow$ radius of liquid interface

$r_2 \rightarrow$ radius of bowl.

if $r_B < r_2 \Rightarrow$ Particle leaves bowl with liquid

$r_B = r_2 \Rightarrow$ deposited on bowl wall & removed from the liquid



In Stokes' law region,
terminal velocity at r
(If particle settles in Stokes' law
region)

$$u_t = \frac{\omega^2 r (R_p - R) D_p^2}{18\mu}$$

$$u_t = \frac{dr}{dt}$$

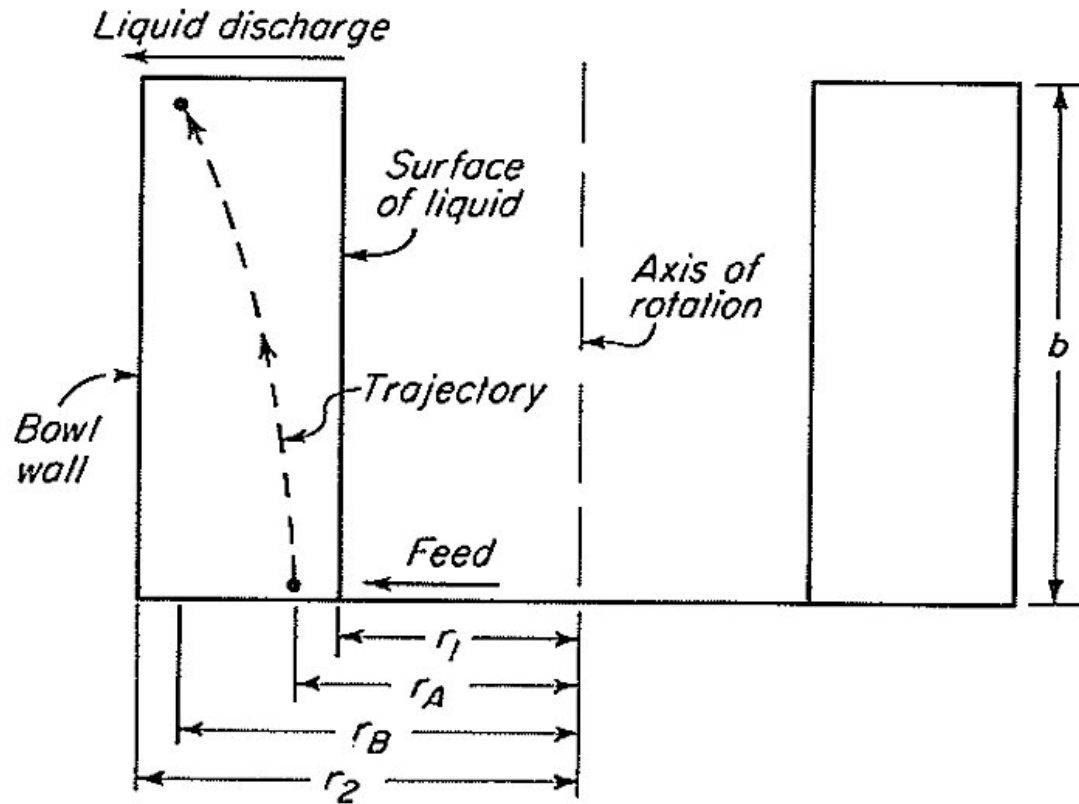
$$dt = \frac{18\mu}{\omega^2 (R_p - R) D_p^2} \frac{dr}{r}$$

Integrate

Limits: $r = r_A$ at $t = 0$

$r = r_B$ at $t = t_T$

$$t_T = \frac{18\mu}{\omega^2 (R_p - R) D_p^2} \ln \left[\frac{r_B}{r_A} \right]$$



$$t_T = \frac{\text{Volume of liquid in bowl}}{\text{Volumetric flow rate}}$$

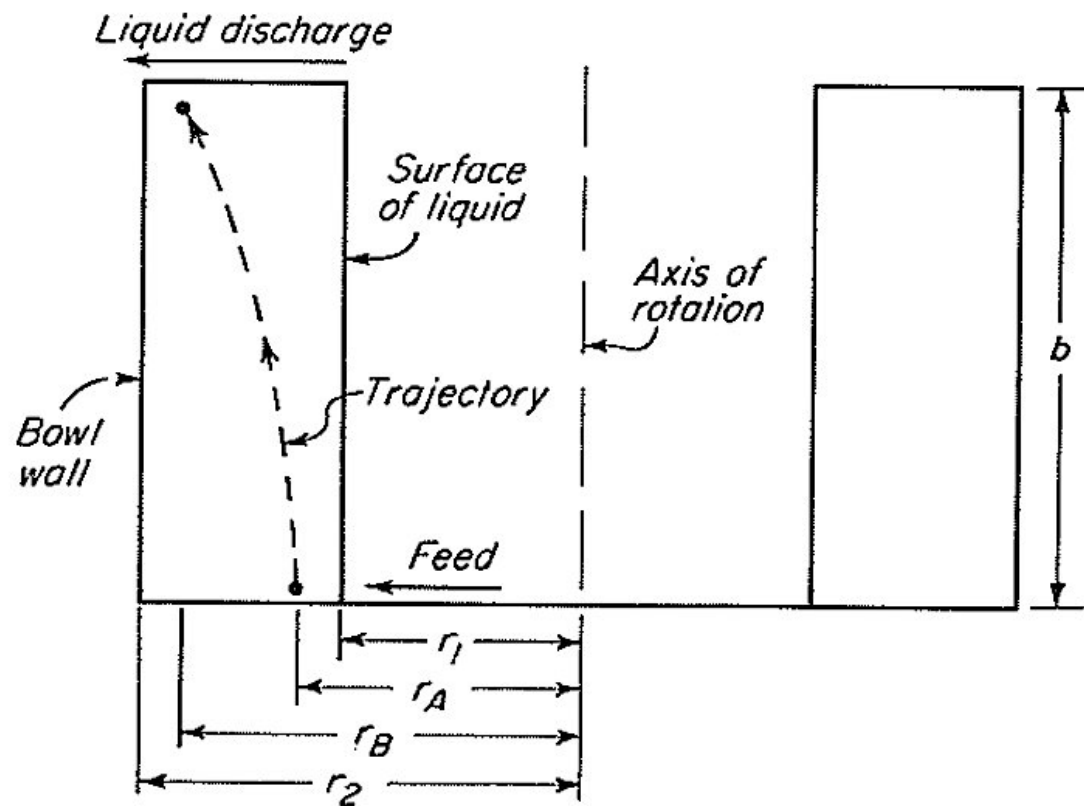
$$= \frac{V}{Q} = \frac{\pi b (r_2^2 - r_1^2)}{Q}$$

$$\frac{\pi b (r_2^2 - r_1^2)}{Q} = \frac{18\mu}{\omega^2 (r_p - r) D_p^2} \ln \left[\frac{r_B}{r_A} \right]$$

$$Q = \frac{\pi b \omega^2 (r_p - r) D_p^2}{18\mu} \frac{r_2^2 - r_1^2}{\ln \left(\frac{r_B}{r_A} \right)}$$

Eqn for Vol flow rate

Cut Point $\rightarrow$ Diameter of the Particle, which reaches one half the distance b/w r_1 & r_2



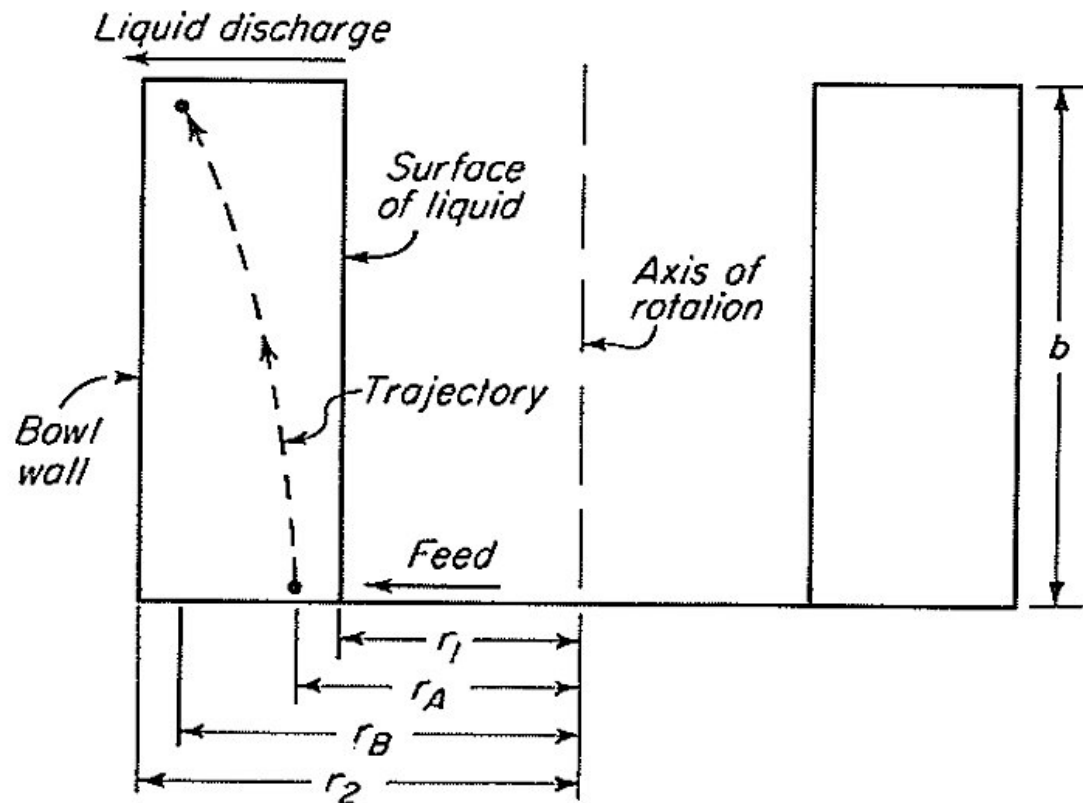
So, $D_{pc} \Rightarrow$ cut diameter a particle of this size moves, $y = \frac{r_1 - r_2}{2}$, during settling time.

If particle of size " D_{pc} " is to be removed, it should reach bowl wall in available time.

Hence, $r_B = r_2$

$$r_A = \frac{r_1 + r_2}{2}$$

$$L_c = \frac{\pi b \omega^2 (R_p - \rho) D_p^2}{18 \mu} \cdot \frac{r_2^2 - r_1^2}{\ln \left[\frac{2r_2}{r_1 + r_2} \right]}$$



$Q_c \Rightarrow$ Vol. flow rate corr. to
Cut Point + Diameter

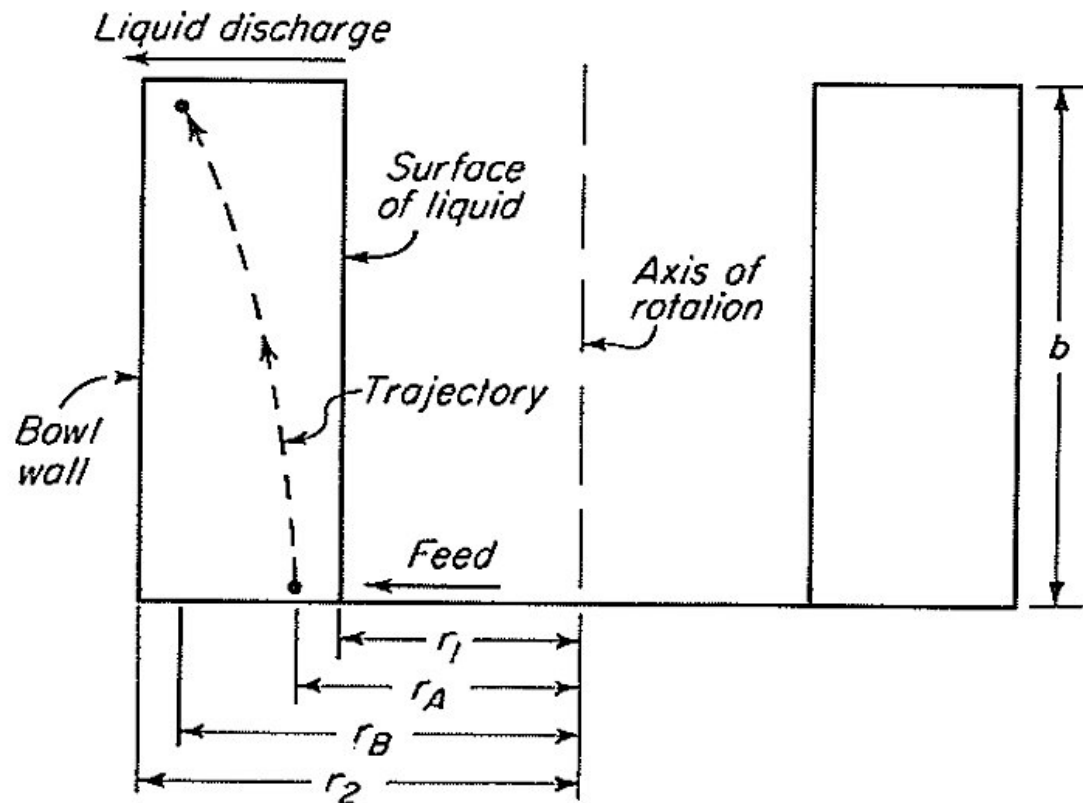
$\Downarrow$
Particles above D_{pc} removed
at this flow rate

Particle $< D_{pc}$ stays in liquid.

When $r_1 \approx r_2$ (thickness of
liquid layer is very small)
Settling Velocity is Constant,

$$u_t = \frac{D_p^2 (\rho_p - \rho) \omega^2 r_2}{18\mu}$$

Thickness of liquid layer $\rightarrow \delta$
Settling distance for Particles
of $D_{pc} \rightarrow \frac{\delta}{2}$



Then
$$u_t = \frac{S/2}{t_T} = \frac{S}{2t_T}$$

$$t_T = \frac{V}{q_c}$$

$$u_t = \frac{S q_c}{2V}$$

$$q_c = \frac{2V u_t}{S}$$

$$q_c = \frac{2V D_p^2 (\rho_p - \rho) \omega^2 r_2}{18\mu s}$$

- Volumetric flow rate:

$$q_c = \frac{\pi b \omega^2 (\rho_p - \rho) D_{pc}^2}{18\mu} \frac{r_2^2 - r_1^2}{\ln [2r_2 / (r_1 + r_2)]}$$

- Volumetric flow rate when liquid is thin

$$q_c = \frac{2Vu_t}{s} = \frac{2VD_p^2(\rho_p - \rho)\omega^2 r_2}{18\mu s}$$

Scale up

For Industrial Centrifuges,
Eqn for flow rate of liquid
with thin layer is modified.

$$Q_c = \frac{2VD_p^2(R_p - R) \omega^2 r_2}{18\mu S}$$

$r_2 \rightarrow r_e$
 $S \rightarrow S_e$ } Avg Values of r & S
for type of Centrifuge
under Consideration.

$$Q_c = \frac{2V\omega^2 r_e}{g_s} \frac{D_p^2 (R_p - R) g}{18\mu}$$
$$= 2 \sum u_g$$

$\sum \Rightarrow$ Characteristics of Centrifuge
 $u_g \Rightarrow$ terminal Settling Vel. under
gravity Settling Conditions.

$\Sigma \Rightarrow$ Cross Sectional Area of
a gravity settling tank
of the same separation
Capacity as Centrifuge.

Sigma value and scale up



- The below equation is used (r_2 and s are replaced by average values r_e and s_e):

$$q_c = \frac{2V\omega^2 r_e D_p^2 (\rho_p - \rho)g}{gs \cdot 18\mu}$$
$$= 2 \Sigma u_g$$

- Sigma value is a characteristic of the centrifuge and u_g is the terminal settling velocity of the particle under gravity.
- Sigma is the cross-sectional area of a gravity settling tank of the same separation capacity as the centrifuge.

TABLE 30.5
Characteristics of sedimenting centrifuges^{29b,38}

Type	Bowl diameter, in.	Speed, r/min	Σ value, $\text{ft}^2 \times 10^{-4}$
Tubular	4.125	15,000	2.7
Disk	9.5	6,500	21.5
	13.7	4,650	39.3
	19.5	4,240	105
Helical conveyor	14	4,000	1.34
	25	3,000	6.1
Axial-flow conveyor			
	No vanes	2,600	4.05
	96 vanes	2,600	12.7

Problem on Centrifugal Sedimentation



- What is the capacity in cubic meters per hour of a clarifying centrifuge operating under the following conditions?
 - Diameter of bowl, 600 mm
 - Thickness of liquid layer, 75 mm
 - Depth of bowl, 400 mm
 - Speed 1200 r/min
 - Specific gravity of liquid 1.2
 - Specific gravity of solid, 1.6
 - Viscosity of liquid 2 cP
 - Cut size of particles, $30\ \mu\text{m}$

$$q_c = \frac{\pi b \omega^2 (\rho_p - \rho) D_{pc}^2}{18\mu} \frac{r_2^2 - r_1^2}{\ln [2r_2/(r_1 + r_2)]}$$

Thank you